# A comment on "They Never had a Chance: Unequal Opportunities and Fair Redistributions"*

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November 11, 2026

**Abstract**

We reassess Dong et al. (2025), henceforth DHL, on redistribution under unequal opportunities. Using the authors' replication package, we rerun every do-file on the raw Qualtrics exports and reproduce all reported figures and tables to the printed digits. The re-analysis confirms DHL's four descriptive claims: spectators redistribute more when outcomes derive from unequal opportunity, and the ordering Luck > Random-Education > Random-Employment > Merit holds throughout.

We also inspect the preregistered matching protocol versus the realized deterministic pairing. In a robustness reproduction, we treat missing demographic and belief responses as missing instead of assigning them a value (the original code classes income non-respondents as high income and fills missing beliefs with zeros), use alternative cutoffs for high education and high income, and tighten the speeding filters, which removes 17% of the main spectator sample and 7% of the follow-up sample. Applied jointly, these changes leave the magnitude and statistical significance of the main descriptive pattern unchanged.

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## 1 Introduction

DHL investigate spectators' fairness preferences when observed performance differences can stem from unequal opportunity. Spectators choose redistributions between a relatively advantaged and a relatively disadvantaged worker across conditions where differences originate from random disparities in education or employment (unequal-opportunity conditions) versus benchmark cases. The main descriptive claim is that spectators shift resources toward the disadvantaged when disparities reflect unequal opportunity.

This descriptive pattern intersects with a long debate about merit and desert: when outcomes reflect advantage rather than ability or effort, redistribution can be perceived as restoring fairness rather than dulling incentives. As already captured in Young (1958)'s classic *The Rise of the Meritocracy*, relabeling advantage as virtue can naturalize unequal starting points. DHL's design lets spectators condition transfers precisely on that boundary between opportunity and effort.

The study relies on a Qualtrics-hosted online experiment fielded between January and April 2022. DHL recruited 842 workers to perform the real-effort tasks that generate redistribution scenarios and, in parallel, English-language spectator cohorts (1,708 respondents for the main redistribution task plus 992 for the Vary modules) who observe the outcomes and choose transfers.

In this report for the Institute for Replication (I4R; Brodeur *et* al. 2024), we (i) carefully evaluate computational reproducibility from raw data using the authors' Stata code, (ii) re-do the data cleaning and analysis in Python as a cross-software check, and (iii) run a robustness exercise where we exclude missing responses that had been coded as zeros (or, in the case of income, as high income) and further redefine the cut-off levels for key binary variables. Overall, the main results reproduce cleanly and the robustness analyses do not alter the qualitative claims.

## 2 Computational Reproducibility

For this report, we use the updated replication package available at `https://doi.org/10.5281/zenodo.16317200`, Version 3. The archive contains (i) raw worker and spectator data, (ii) processed Stata datasets, (iii) a master script (`0PathSetup.do`)

that sequentially calls the cleaning and analysis routines, and (iv) the graph/table outputs embedded in the published PDF. The README clearly documents the workflow and required Stata version.

We execute the master do-file from the project root after fixing `version 16.0` and the recommended random seed; every script runs without modification. The cleaning scripts recreate `spectator-main.dta` and `spectator-vary.dta` from the raw Qualtrics downloads, and the analysis scripts reproduce DHL's Figures 1–3 and all numbered tables to the printed digits. Table 1 in the Appendix summarizes our inventory of the replication package and confirms that each component required for a raw-data reproduction is present, while the Appendix compiles every replicated table (Tables 2–11) generated by the Stata pipeline. Our annotated code, along with the modified figures and tables, accompany this report in the public archive at https://doi.org/10.5281/zenodo.17727800.

**Cross-software reproduction in Python.** To assess *recreate reproducibility*, we import the raw data into Python, rebuild the cleaning steps (re-weighting of redistributive choices, trimming of low-quality responses, and treatment-specific aggregation), and recreate Figures 1–3. Point estimates match their Stata counterparts up to rounding, and the ordering of treatments and confidence intervals is identical.

## 3 Discrepancies Between Pre-analysis Plan and Article

The preregistration (https://doi.org/10.1257/rct.8474-2.0) described "random ex-post" pairing of workers in the unequal-opportunity conditions. In the realized implementation, pairs were matched deterministically so that the higher-opportunity worker also scored higher and earned \$6 in Random-Education and Random-Employment. Consequently, a disadvantaged worker never "won." Spectators observed the realized scores and opportunity labels, but they were not told that an underdog win was impossible. The published article (Sec. 1.2.2, footnote 9) reports this deviation. While deterministic pairing could, in principle, make observers more confident that low scores stem from luck rather than effort, we classify this choice as a minor deviation from the pre-analysis plan.

## 4 Robustness reproduction

**Variable abbreviations in Stata.** With Stata's default settings, all code executes from the master file `0PathSetup.do` once the replication package is set as current working directory. To ensure compatibility we fix Stata `version 16.0` and, as a first check, rerun the master script after changing the default to `set varabbrev off`. We find three abbreviated calls (`scoreemployment`, `finish`, `female1`). Because the abbreviations are unique, results remain identical, yet we strongly recommend turning the option off permanently to avoid silent mis-specification.

**Missing demographics.** The original scripts build the binary indicators (`female1`, `highEdu`, `highIncome`, `conservative`) from logical expressions that never return a missing value, so a respondent without a valid answer is silently placed in one of the two groups. Which group differs by variable. For gender, "Other or prefer not to say" is coded as zero and thus grouped with men. For income, "I prefer not to answer" is first set to missing, but the indicator is then generated as `income1 > 50000`; because Stata treats a missing value as larger than any number, these respondents are classed as *high* income, not as zero. This is a coding error rather than a mere convention. Our cleaning log confirms the mechanism: before our re-coding, all 481 rows with missing income in the raw file carry `highIncome` = 1. For education and political affiliation, the only missing values are blank answers. Each of the two questions has 447 blank answers among the 1,708 rows of the raw main-experiment file, matching the 437 respondents who failed the comprehension quiz plus the 10 who did not complete the survey in DHL's documentation, and all of these rows are dropped before the analysis.

We re-code the indicators to keep true missings, treating "Other or prefer not to say" as missing for gender, and recompute the splits. In our adjusted analysis sample of 666 spectators (see "Further stress-tests" below), this removes 4 spectators from the gender split and 23 from the income split. It changes nothing in the education and political splits, because no spectator in the analysis sample has a blank education or party answer (in Appendix Table 12, the four panels sum to $N = 662$, 643, 666 and 666, respectively).

The right-hand panel of Figure 1 comes from the same pipeline run as the

alternative cutoffs and the stricter duration filter described below, so it reflects all of these modifications jointly rather than the re-coding of missing values alone. The Political panel is essentially unchanged: in Random-Education, non-conservatives redistribute 0.37 and conservatives 0.29 in both columns, and no other bar in this panel moves by more than 0.01. The visible changes are in the Education panel, where the two columns do not compare like with like: the lowered high-education threshold leaves only 17–20 low-education spectators per treatment, and their mean share falls from 0.35 to 0.28 in Random-Education and from 0.22 to 0.18 in Merit. The Income panel, which combines the re-coded missings with the higher $70,000 threshold, moves by up to 0.03, and the Gender panel by at most 0.01. Appendix Table 12 lists the subgroup means and confidence intervals that underlie the right-hand panel of Figure 1 for ease of reference.

**Missing beliefs.** For the belief outcomes (perceived role of education or employment in explaining outcomes), missing responses were filled with zeros before computing means. We replicate the graphs while excluding missing responses entirely. This change was not run in isolation: the right-hand panels of Figures 2 and 3 come from the single pipeline in which all modifications of this section were applied together. Besides the exclusion of missing beliefs, they therefore reflect the stricter 300-second duration filter described below, which reduces the Vary samples from 225/210/211 to 207/193/198 spectators (Vary-Probability/Vary-Education/Vary-Employment) and the meritocratic subsamples from 182/86/58 to 170/82/56; the demographic re-codes and cutoffs do not enter these figures. As shown in Figures 2 and 3, under these joint changes the means shift only marginally – most notably a slight attenuation in the Vary-Education panel – while the ordering of treatments and confidence intervals remains intact. The differences between the left and right panels cannot be attributed to the treatment of missing beliefs alone. Appendix Tables 13 and 14 tabulate the scenario-level means and 95% confidence intervals that correspond to the right-hand panels of Figures 2 and 3.

**Further stress-tests.** We further investigate whether the exclusion criteria and specific cut-off choices affect the results. In particular, we re-define what qualifies as high education, high income, as well as too-fast responses. High income is set at above $70,000 (instead of $50,000) and high education at "some college education

or a two-year degree" (instead of DHL's four-year threshold). To guard against low-effort responses, we tighten the duration filters to 200 and 300 seconds (from DHL's 120 and 180 seconds) when constructing `spectator-main.dta` and `spectator-vary.dta`. This reduces the sample substantially: the four main treatments fall from 806 spectators (201/201/201/203 in DHL's Table 2) to 666 (156/169/170/171 in our Table 2), a cut of 17%, and the Vary modules fall from 225/210/211 to 207/193/198 spectators, a cut of 7%.

The missing-data re-codes, the alternative cutoffs, and the stricter duration filters were all applied together in one pipeline rather than one at a time. Every right-hand panel in Figures 1, 2 and 3 and every appendix table from Table 2 onwards therefore reflects all three sets of changes at once, and a difference from DHL's results cannot be attributed to any single one of them. Tables 2–11 report key regression results from this joint specification and show that the treatment contrasts remain qualitatively unchanged.

Across these checks, DHL's main descriptive claim—that spectators channel more redistribution toward the disadvantaged in unequal-opportunity conditions—remains intact. The treatment means move by less than 0.02 on the 0–1 redistribution scale in every exercise, and the Luck > Merit and Random-Education > Random-Employment contrasts remain significant at $p < 0.01$.

## 5 Conclusion

Our replication reproduces DHL's main figures and balance table from raw data using the provided Stata code and an independent Python re-implementation. Following I4R's taxonomy, we complete: (i) *computational reproducibility* (rerun authors' code on raw inputs); (ii) *recreate reproducibility* (independent cross-software implementation); and a basic (iii) *robustness reproducibility* (Section 4). Across the four descriptive claims emphasized by DHL – Luck > Merit; Random-Education between Luck and Merit; Random-Employment between Luck and Merit; Random-Education > Random-Employment – we reproduce 4/4 (100%). All coefficients, standard errors, and p-values match the published tables within rounding tolerance. The preregistered claims remain valid after re-coding missing demographic

items and stress-testing alternative cutoffs. We therefore conclude that DHL's central descriptive message is robust.

## References


Brodeur *et* al.: 2024, Mass Reproducibility and Replicability: A New Hope, *I4R Discussion Paper Series* .

Dong, L., Huang, L. and Lien, J. W.: 2025, 'They Never had a Chance': Unequal opportunities and fair redistributions, *The Economic Journal* **135**(667), 914–942.

Young, M.: 1958, *The Rise of the Meritocracy, 1870–2033: An Essay on Education and Equality*, Thames and Hudson, London.

## Figures

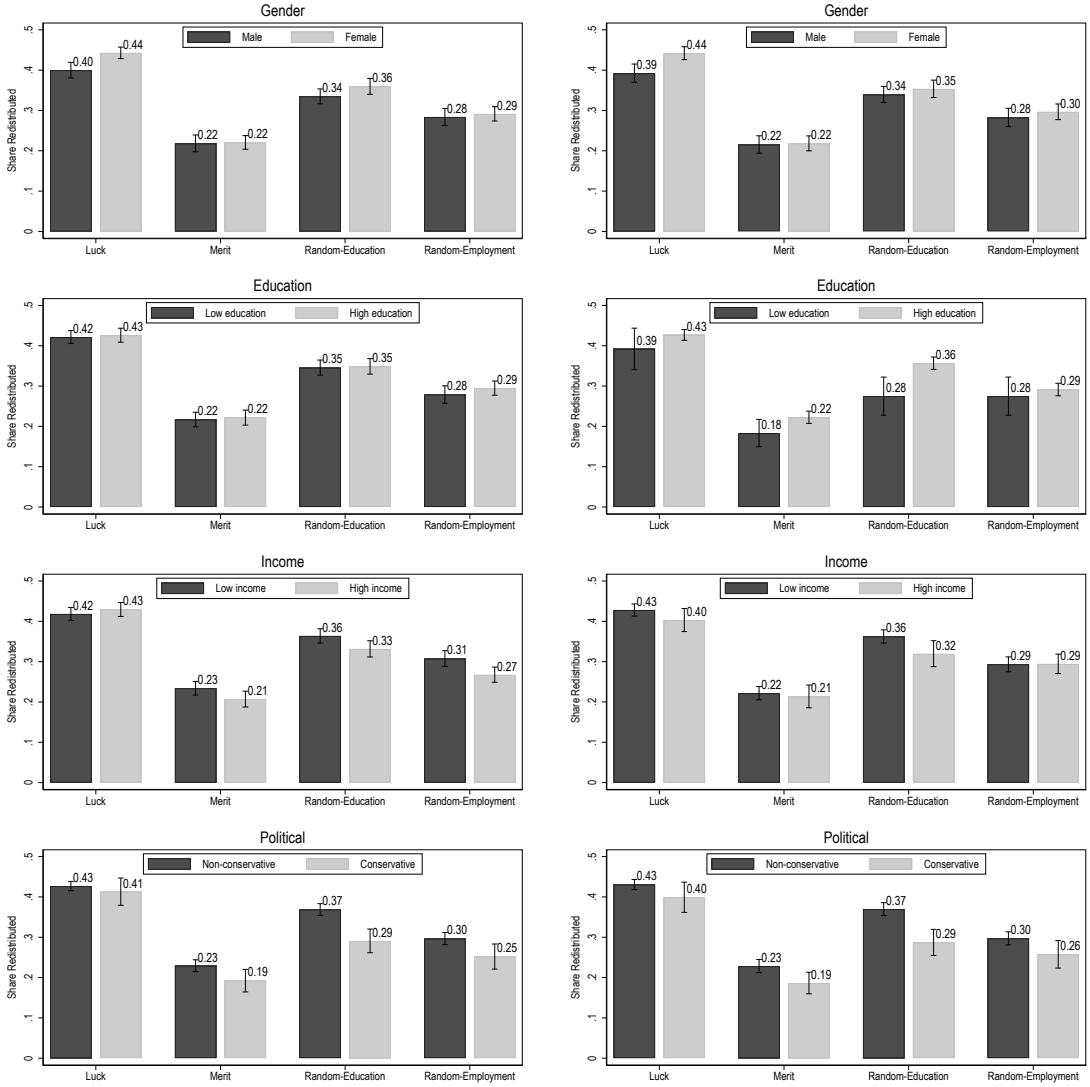



Figure 1: Share redistributed to the disadvantaged worker by treatment (Luck, Merit, Random-Education, Random-Employment) and spectator characteristics: gender, education, income, and political affiliation. Bars depict subgroup means with 95% confidence intervals. The left column reproduces DHL Figure A2; the right column re-creates it with our adjusted spectator sample.

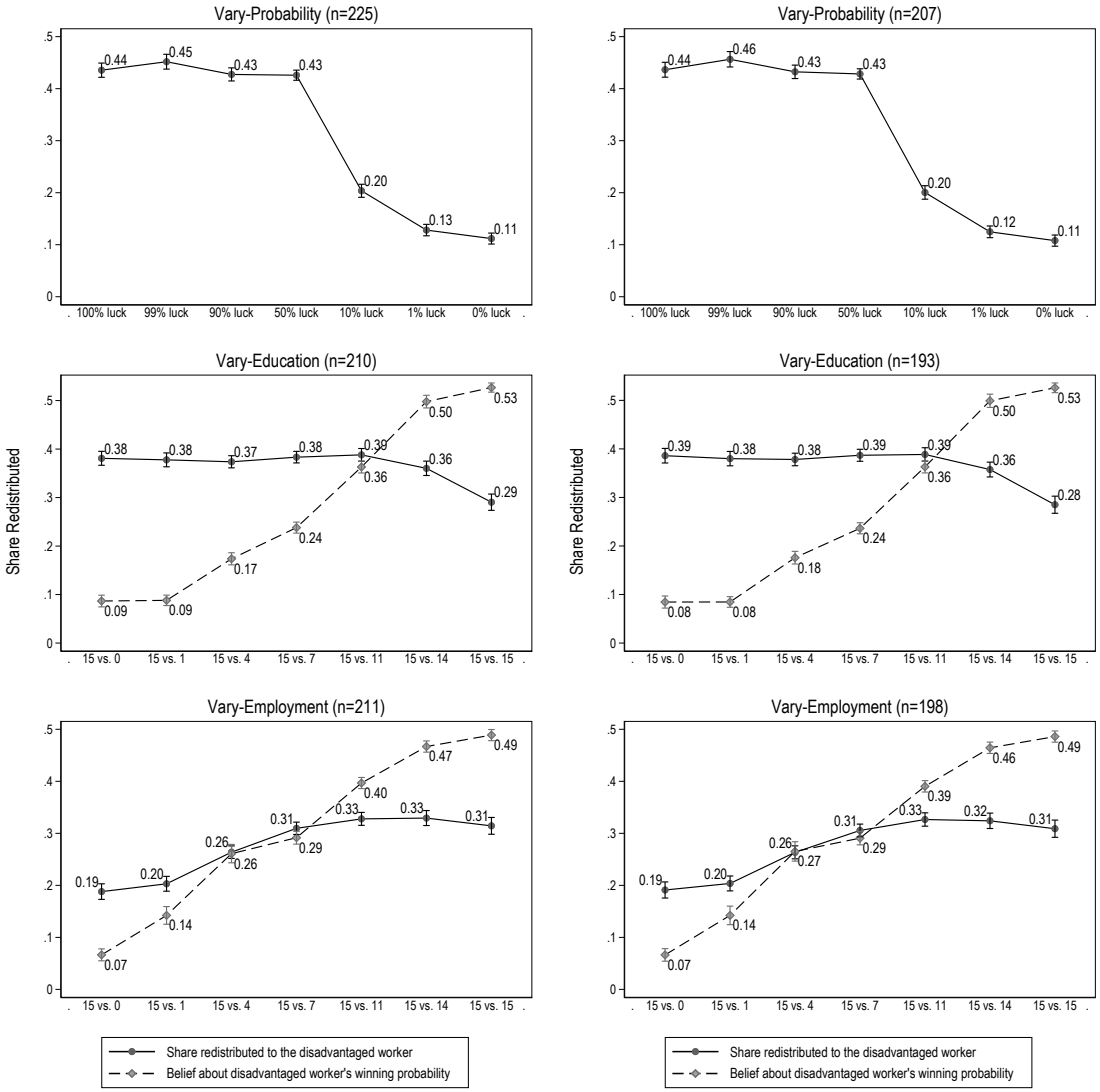



Figure 2: Average share redistributed to the disadvantaged worker (dark line) and spectators' beliefs about the disadvantaged worker's winning probability (light line) across the Vary-Probability, Vary-Education, and Vary-Employment tasks. The left panels reproduce DHL Figure 2; the right panels re-create the same statistic from our jointly modified pipeline, which excludes missing beliefs and applies the stricter duration filter (Section 4).

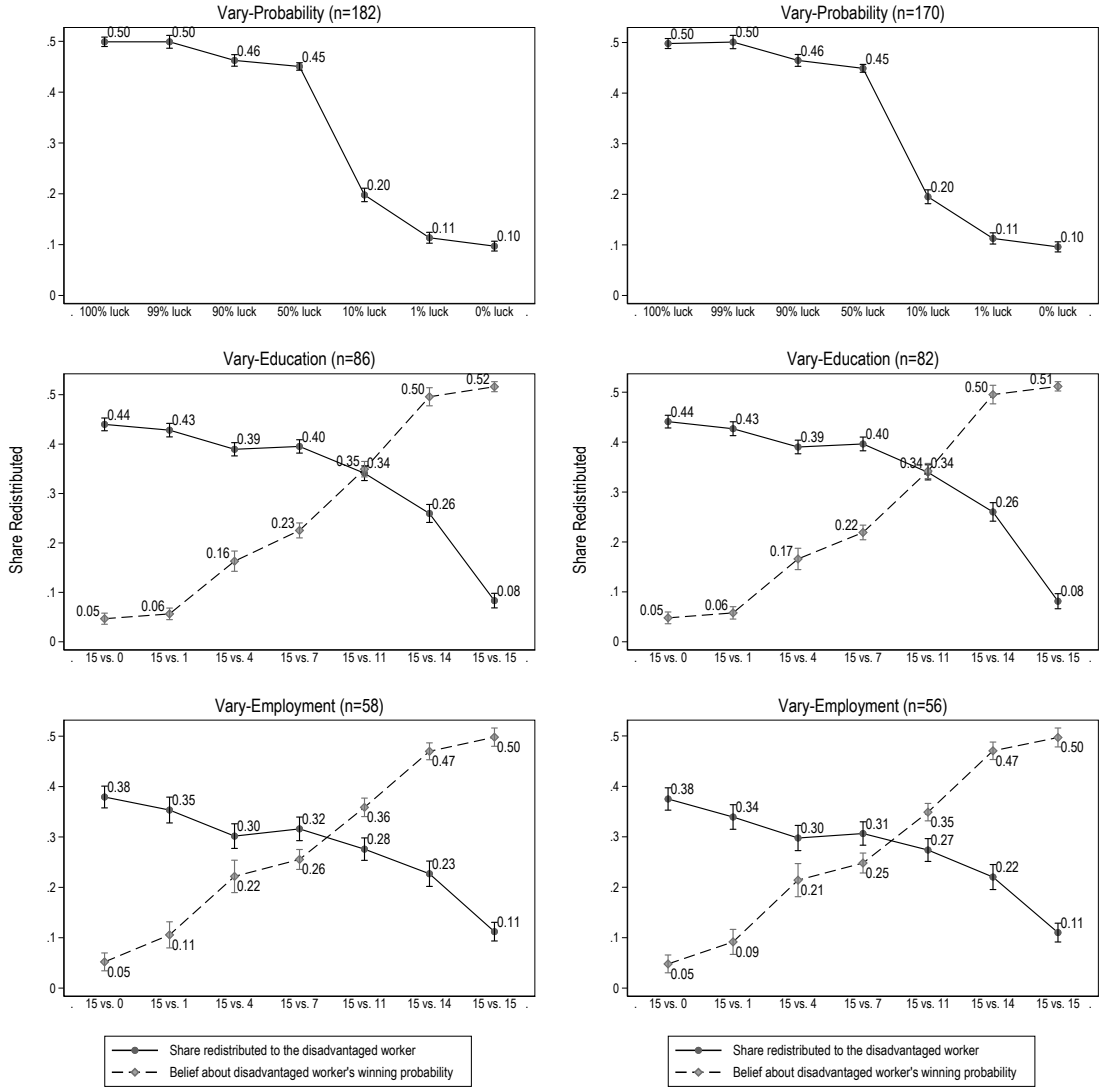



Figure 3: Subsample of "meritocratic" spectators (those who reallocate little when the advantaged worker fully deserves the outcome) in the Vary-Probability, Vary-Education, and Vary-Employment tasks. Each panel plots the mean share redistributed to the disadvantaged worker with 95% confidence intervals. Left panels replicate DHL Figure 3, while the right panels show the same calculation from our jointly modified pipeline, which excludes missing beliefs and applies the stricter duration filter (Section 4).

## 6 Appendix

Table 1: Replication Package Contents and Reproducibility

| **Replication Package Item** | **Fully** | **Partial** | **No** |
|---|---|---|---|
| Raw data provided | ✓ | | |
| Analysis data provided | ✓ | | |
| Cleaning code provided | ✓ | | |
| Analysis code provided | ✓ | | |
| Reproducible from raw data | ✓ | | |
| Reproducible from analysis data | ✓ | | |

*Notes*: This table summarizes the replication package contents available at https://doi.org/10.5281/zenodo.16317200.

| | Luck | Merit | Random-Education | Random-Employment |
|---|---|---|---|---|
| Female (share) | .5974026 | .5952381 | .5294118 | .5352941 |
| Average age (years) | 46.97436 | 47.14201 | 45.74118 | 45.59064 |
| High education (share) | .8910256 | .8816568 | .8823529 | .8830409 |
| Average income (USD) | 57648.03 | 58185.98 | 57730.06 | 58597.56 |
| Republican (share) | .2435897 | .2485207 | .2764706 | .1929825 |
| Observations | 156 | 169 | 170 | 171 |

Table 2: Spectator demographics and balances by treatment (Luck, Merit, Random-Education, and Random-Employment). This replicates DHL Table 2 using the cleaned replication sample.

| | Luck (Vary-Probability)<br>Share redistributed | Random-Education (Vary module)<br>Share redistributed | Random-Employment (Vary module)<br>Share redistributed |
|---|---|---|---|
| alpha | 2.523*** | 3.534*** | 2.095*** |
| | (0.060) | (0.143) | (0.078) |
| /Residual lnsigma | -2.019*** | -2.051*** | -1.791*** |
| | (0.020) | (0.030) | (0.036) |
| $N$ | 1190 | 574 | 392 |

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 3: Structural estimates of the curvature parameter $\alpha$ of the spectators' loss function, following DHL Table 3 (their Section 3.2.2) and re-estimated on our adjusted sample. As in DHL's code, $\alpha$ is estimated for meritocratic spectators by maximum likelihood with Stata's non-linear mixed-effects command `menl`, separately for each task. The three columns correspond to the Vary-Probability, Vary-Education, and Vary-Employment tasks.

| | Random-Education (share) | | | Random-Employment (share) | | |
|---|---|---|---|---|---|---|
| | Mean | Std. Dev. | N | Mean | Std. Dev. | N |
| Similar effort | .4243986 | .1382359 | 97 | .2956989 | .1861385 | 31 |
| Doubt effort in either task | .3315972 | .2030395 | 96 | .257485 | .1732694 | 167 |
| Doubt effort in reading task | .3240741 | .2128297 | 72 | .2395833 | .1670884 | 112 |
| Doubt effort in answering task | .3351852 | .2052593 | 90 | .2592593 | .1742578 | 162 |

Table 4: Perceived effort differences (mean, standard deviation, and observations) across Vary-Education and Vary-Employment prompts. Each row corresponds to DHL Table 4 belief descriptors.

| | (1)<br>Share redistributed | (2)<br>Share redistributed |
|---|---|---|
| Merit | -0.205*** | -0.201*** |
| | (0.019) | (0.019) |
| Random-Education | -0.076*** | -0.066*** |
| | (0.020) | (0.020) |
| Random-Employment | -0.134*** | -0.127*** |
| | (0.020) | (0.020) |
| Female | | 0.018 |
| | | (0.015) |
| Age (years) | | 0.001 |
| | | (0.000) |
| High education | | 0.040* |
| | | (0.024) |
| High income | | -0.022 |
| | | (0.016) |
| Republican | | -0.049*** |
| | | (0.018) |
| Constant | 0.423*** | 0.355*** |
| | (0.013) | (0.035) |
| *N* | 666 | 641 |

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: DHL table A1: OLS regression results on share redistributed by spectators. The Luck treatment serves as the reference category. "High income" is an indicator variable for having yearly income higher than $70,000 (instead of DHL's $50,000 categorization). "High education" is an indicator variable for having at least some college education (instead of DHL's definition as a 4-year college education or higher). "Conservative" is an indicator variable for having selected Republican as their political party/stance most typically supported (cf. DHL table A1).

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Merit | -0.177*** | -0.209*** | -0.206*** | -0.202*** |
| | (0.031) | (0.060) | (0.022) | (0.021) |
| Random-Education | -0.053* | -0.117* | -0.066*** | -0.061*** |
| | (0.030) | (0.068) | (0.022) | (0.020) |
| Random-Employment | -0.110*** | -0.117* | -0.135*** | -0.134*** |
| | (0.032) | (0.068) | (0.024) | (0.021) |
| Merit | | | | |
| times Female | -0.047 | | | |
| | (0.040) | | | |
| Random-Education | | | | |
| times Female | -0.035 | | | |
| | (0.040) | | | |
| Random-Employment | | | | |
| times Female | -0.036 | | | |
| | (0.041) | | | |
| Female | 0.050* | | | |
| | (0.028) | | | |
| Merit | | | | |
| times High education | | 0.005 | | |
| | | (0.064) | | |
| Random-Education | | | | |
| times High education | | 0.047 | | |
| | | (0.071) | | |
| Random-Employment | | | | |
| times High education | | -0.018 | | |
| | | (0.071) | | |
| High education | | 0.035 | | |
| | | (0.052) | | |
| Merit | | | | |
| times High income | | | 0.017 | |
| | | | (0.046) | |
| Random-Education | | | | |
| times High income | | | -0.018 | |
| | | | (0.048) | |
| Random-Employment | | | | |
| times High income | | | 0.026 | |
| | | | (0.044) | |
| High income | | | -0.025 | |
| | | | (0.032) | |
| Merit | | | | |
| times Republican | | | | -0.010 |
| | | | | (0.050) |
| Random-Education | | | | |
| times Republican | | | | -0.051 |
| | | | | (0.053) |
| Random-Employment | | | | |
| times Republican | | | | -0.008 |
| | | | | (0.054) |
| Republican | | | | -0.032 |
| | | | | (0.039) |
| Constant | 0.392*** | 0.392*** | 0.428*** | 0.431*** |
| | (0.023) | (0.050) | (0.015) | (0.013) |
| *N* | 662 | 666 | 643 | 666 |

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: OLS estimates of redistribution shares on treatment indicators and their interactions with spectator demographics using the cleaned replication sample. Columns (1)–(4) mirror DHL Table A4 by interacting treatments with gender, education, income, and political affiliation, respectively, with Luck as the reference category.

| | Vary-Probability | Vary-Education | Vary-Employment |
|---|---|---|---|
| Female (share) | .4417476 | .4315789 | .4093264 |
| Average age (years) | 38.343 | 40.5285 | 39.35354 |
| High education (share) | .8792271 | .9119171 | .8686869 |
| Average income (USD) | 63453.82 | 62464.76 | 63055.77 |
| Republican (share) | .2459016 | .2263158 | .2081633 |
| Observations | 244 | 190 | 196 |

Table 7: Spectator demographics (gender, age, education, income, and ideology) for the Vary-Probability, Vary-Education, and Vary-Employment belief modules, replicating DHL Table A6 on the adjusted sample.

| | (1) | (2) | (3) |
|---|---|---|---|
| 99% luck / 15 vs. 1 | 0.020 | -0.006 | 0.013 |
| | (0.013) | (0.008) | (0.008) |
| 90% luck / 15 vs. 4 | -0.004 | -0.008 | 0.072*** |
| | (0.013) | (0.011) | (0.015) |
| 50% luck / 15 vs. 7 | -0.008 | 0.001 | 0.114*** |
| | (0.013) | (0.010) | (0.016) |
| 10% luck / 15 vs. 11 | -0.236*** | 0.003 | 0.136*** |
| | (0.018) | (0.016) | (0.019) |
| 1% luck / 15 vs. 14 | -0.312*** | -0.028 | 0.133*** |
| | (0.017) | (0.019) | (0.021) |
| 0% luck / 15 vs. 15 | -0.329*** | -0.101*** | 0.118*** |
| | (0.017) | (0.022) | (0.024) |
| Constant | 0.436*** | 0.386*** | 0.191*** |
| | (0.014) | (0.015) | (0.016) |
| $N$ | 1449 | 1351 | 1386 |

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: OLS estimates of the share redistributed on scenario indicators in the follow-up (Vary) experiment, following DHL Table A7 and re-estimated on our adjusted sample. Columns (1)–(3) correspond to the Vary-Probability, Vary-Education, and Vary-Employment tasks and include all spectators in each task. The 100% luck or 15 vs. 0 scenario is the reference category; standard errors are clustered at the spectator level.

| | (1) | (2) | (3) |
|---|---|---|---|
| 99% luck / 15 vs. 1 | 0.022* | -0.010 | 0.002 |
| | (0.013) | (0.006) | (0.010) |
| 90% luck / 15 vs. 4 | -0.002 | -0.035*** | 0.032 |
| | (0.014) | (0.013) | (0.021) |
| 50% luck / 15 vs. 7 | -0.007 | -0.030** | 0.044** |
| | (0.014) | (0.012) | (0.022) |
| 10% luck / 15 vs. 11 | -0.248*** | -0.057*** | 0.037 |
| | (0.018) | (0.022) | (0.025) |
| 1% luck / 15 vs. 14 | -0.327*** | -0.133*** | -0.015 |
| | (0.017) | (0.022) | (0.028) |
| 0% luck / 15 vs. 15 | -0.346*** | -0.295*** | -0.087*** |
| | (0.017) | (0.023) | (0.028) |
| Constant | 0.432*** | 0.365*** | 0.204*** |
| | (0.015) | (0.020) | (0.022) |
| $N$ | 1372 | 693 | 721 |

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9: OLS estimates of the share redistributed on scenario indicators for meritocratic spectators in the follow-up (Vary) experiment, following DHL Table A8 and re-estimated on our adjusted sample. As in DHL's code, the sample is restricted to spectators who give the lower earner less than half in the scenario without uncertainty (0% luck or 15 vs. 15). Columns (1)–(3) correspond to the Vary-Probability, Vary-Education, and Vary-Employment tasks. The 100% luck or 15 vs. 0 scenario is the reference category; standard errors are clustered at the spectator level.

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Education information treatment | 0.037 | 0.034 | | |
| | (0.023) | (0.024) | | |
| Uninformed spectators | -0.033 | -0.040 | -0.007 | -0.010 |
| | (0.025) | (0.025) | (0.026) | (0.027) |
| Female | | 0.006 | | 0.002 |
| | | (0.020) | | (0.020) |
| Age (years) | | -0.000 | | 0.000 |
| | | (0.001) | | (0.001) |
| High education | | 0.051 | | 0.031 |
| | | (0.033) | | (0.032) |
| High income | | -0.027 | | -0.008 |
| | | (0.023) | | (0.021) |
| Republican | | -0.066*** | | -0.028 |
| | | (0.023) | | (0.026) |
| Employment information treatment | | | 0.032 | 0.028 |
| | | | (0.024) | (0.025) |
| Constant | 0.347*** | 0.330*** | 0.289*** | 0.272*** |
| | (0.015) | (0.042) | (0.015) | (0.046) |
| $N$ | 364 | 354 | 372 | 356 |

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10: Information-treatment regressions for the Random-Education and Random-Employment games. Columns (1) and (2) compare informed versus uninformed spectators in the education condition; Columns (3) and (4) repeat the analysis for the employment condition. The dependent variable is the share redistributed to the disadvantaged worker (0–1 scale), and the specifications match DHL Table B1 with and without demographic controls.

| | Merit-Training | Random-Training | Merit-Department | Random-Department | Info-Training | Info-Department |
|---|---|---|---|---|---|---|
| Female | .528169 | .5285714 | .5555556 | .5833333 | .4964029 | .5915493 |
| Age | 40.71831 | 40.09286 | 40.9037 | 38.90972 | 41.43165 | 40.95775 |
| High education | .528169 | .5357143 | .5111111 | .4861111 | .618705 | .5704225 |
| Income | 57659.57 | 50000 | 58825.76 | 61482.14 | 55638.69 | 49708.03 |
| conservative | .2464789 | .2428571 | .2666667 | .2083333 | .2230216 | .2605634 |
| $N$ | 142 | 140 | 135 | 144 | 139 | 142 |

Table 11: Demographics for the Merit/Random training and department subsamples, along with the information treatments, as in DHL Table C1. Columns list mean gender, age, education, income, ideology, and the corresponding sample size.

Table 12: Subgroup means corresponding to Figure 1

| Panel | Treatment | Group | Mean share | 95% CI | N |
|---|---|---|---|---|---|
| Gender | Luck | Male | 0.392 | [0.348, 0.437] | 62 |
| Gender | Luck | Female | 0.442 | [0.411, 0.473] | 92 |
| Gender | Merit | Male | 0.216 | [0.173, 0.258] | 68 |
| Gender | Merit | Female | 0.218 | [0.182, 0.255] | 100 |
| Gender | Random-Education | Male | 0.340 | [0.300, 0.379] | 80 |
| Gender | Random-Education | Female | 0.354 | [0.312, 0.396] | 90 |
| Gender | Random-Employment | Male | 0.283 | [0.238, 0.327] | 79 |
| Gender | Random-Employment | Female | 0.297 | [0.259, 0.335] | 91 |
| Education | Luck | Low education | 0.392 | [0.291, 0.493] | 17 |
| Education | Luck | High education | 0.427 | [0.401, 0.453] | 139 |
| Education | Merit | Low education | 0.183 | [0.117, 0.250] | 20 |
| Education | Merit | High education | 0.223 | [0.193, 0.252] | 149 |
| Education | Random-Education | Low education | 0.275 | [0.182, 0.368] | 20 |
| Education | Random-Education | High education | 0.357 | [0.327, 0.387] | 150 |
| Education | Random-Employment | Low education | 0.275 | [0.182, 0.368] | 20 |
| Education | Random-Employment | High education | 0.291 | [0.261, 0.322] | 151 |
| Income | Luck | Low income | 0.428 | [0.399, 0.458] | 109 |
| Income | Luck | High income | 0.403 | [0.347, 0.459] | 43 |
| Income | Merit | Low income | 0.222 | [0.190, 0.254] | 118 |
| Income | Merit | High income | 0.214 | [0.158, 0.269] | 46 |
| Income | Random-Education | Low income | 0.363 | [0.330, 0.395] | 114 |
| Income | Random-Education | High income | 0.320 | [0.257, 0.383] | 49 |
| Income | Random-Employment | Low income | 0.293 | [0.257, 0.330] | 121 |
| Income | Random-Employment | High income | 0.295 | [0.247, 0.342] | 43 |
| Political affiliation | Luck | Non-conservative | 0.431 | [0.406, 0.455] | 118 |
| Political affiliation | Luck | Conservative | 0.399 | [0.326, 0.473] | 38 |
| Political affiliation | Merit | Non-conservative | 0.228 | [0.196, 0.260] | 127 |
| Political affiliation | Merit | Conservative | 0.187 | [0.134, 0.239] | 42 |
| Political affiliation | Random-Education | Non-conservative | 0.370 | [0.339, 0.401] | 123 |
| Political affiliation | Random-Education | Conservative | 0.287 | [0.224, 0.351] | 47 |
| Political affiliation | Random-Employment | Non-conservative | 0.297 | [0.265, 0.329] | 138 |
| Political affiliation | Random-Employment | Conservative | 0.258 | [0.191, 0.324] | 33 |

*Notes*: Average share redistributed to the disadvantaged worker (0–1 scale) for each treatment and subgroup; confidence intervals mirror the bars in Figure 1.

Table 13: Scenario-level statistics for Figure 2

| Panel | Scenario | Mean share | 95% CI (share) | N (share) | Mean belief | 95% CI (belief) | N (belief) |
|---|---|---|---|---|---|---|---|
| Vary-Probability | 100% luck | 0.436 | [0.408, 0.464] | 207 | – | – | – |
| Vary-Probability | 99% luck | 0.457 | [0.428, 0.486] | 207 | – | – | – |
| Vary-Probability | 90% luck | 0.432 | [0.407, 0.458] | 207 | – | – | – |
| Vary-Probability | 50% luck | 0.428 | [0.409, 0.448] | 207 | – | – | – |
| Vary-Probability | 10% luck | 0.200 | [0.175, 0.226] | 207 | – | – | – |
| Vary-Probability | 1% luck | 0.125 | [0.103, 0.147] | 207 | – | – | – |
| Vary-Probability | 0% luck | 0.108 | [0.087, 0.129] | 207 | – | – | – |
| Vary-Education | 15 vs. 0 | 0.386 | [0.357, 0.415] | 193 | 0.085 | [0.060, 0.109] | 190 |
| Vary-Education | 15 vs. 1 | 0.380 | [0.351, 0.409] | 193 | 0.085 | [0.063, 0.106] | 191 |
| Vary-Education | 15 vs. 4 | 0.378 | [0.353, 0.404] | 193 | 0.176 | [0.150, 0.201] | 191 |
| Vary-Education | 15 vs. 7 | 0.387 | [0.363, 0.411] | 193 | 0.236 | [0.214, 0.259] | 191 |
| Vary-Education | 15 vs. 11 | 0.389 | [0.362, 0.416] | 193 | 0.363 | [0.339, 0.387] | 190 |
| Vary-Education | 15 vs. 14 | 0.358 | [0.328, 0.387] | 193 | 0.499 | [0.472, 0.526] | 193 |
| Vary-Education | 15 vs. 15 | 0.285 | [0.250, 0.319] | 193 | 0.526 | [0.506, 0.545] | 193 |
| Vary-Employment | 15 vs. 0 | 0.191 | [0.161, 0.221] | 198 | 0.066 | [0.043, 0.090] | 193 |
| Vary-Employment | 15 vs. 1 | 0.204 | [0.176, 0.232] | 198 | 0.142 | [0.107, 0.177] | 197 |
| Vary-Employment | 15 vs. 4 | 0.263 | [0.239, 0.288] | 198 | 0.265 | [0.229, 0.302] | 196 |
| Vary-Employment | 15 vs. 7 | 0.306 | [0.282, 0.329] | 198 | 0.291 | [0.265, 0.316] | 196 |
| Vary-Employment | 15 vs. 11 | 0.327 | [0.301, 0.352] | 198 | 0.390 | [0.369, 0.412] | 198 |
| Vary-Employment | 15 vs. 14 | 0.324 | [0.295, 0.353] | 198 | 0.464 | [0.443, 0.486] | 197 |
| Vary-Employment | 15 vs. 15 | 0.309 | [0.276, 0.341] | 198 | 0.486 | [0.465, 0.507] | 197 |

*Notes*: Panel sample sizes: Vary-Probability: n=207; Vary-Education: n=193; Vary-Employment: n=198. Beliefs report the perceived probability that the disadvantaged worker wins.

Table 14: Scenario-level statistics for Figure 3

| Panel | Scenario | Mean share | 95% CI (share) | N (share) | Mean belief | 95% CI (belief) | N (belief) |
|---|---|---|---|---|---|---|---|
| Vary-Probability | 100% luck | 0.498 | [0.479, 0.517] | 170 | – | – | – |
| Vary-Probability | 99% luck | 0.501 | [0.475, 0.527] | 170 | – | – | – |
| Vary-Probability | 90% luck | 0.465 | [0.442, 0.488] | 170 | – | – | – |
| Vary-Probability | 50% luck | 0.449 | [0.434, 0.464] | 170 | – | – | – |
| Vary-Probability | 10% luck | 0.195 | [0.168, 0.222] | 170 | – | – | – |
| Vary-Probability | 1% luck | 0.113 | [0.091, 0.134] | 170 | – | – | – |
| Vary-Probability | 0% luck | 0.096 | [0.076, 0.116] | 170 | – | – | – |
| Vary-Education | 15 vs. 0 | 0.441 | [0.416, 0.466] | 82 | 0.048 | [0.025, 0.071] | 82 |
| Vary-Education | 15 vs. 1 | 0.427 | [0.400, 0.454] | 82 | 0.058 | [0.034, 0.082] | 81 |
| Vary-Education | 15 vs. 4 | 0.390 | [0.364, 0.417] | 82 | 0.166 | [0.124, 0.208] | 82 |
| Vary-Education | 15 vs. 7 | 0.396 | [0.369, 0.423] | 82 | 0.219 | [0.190, 0.248] | 82 |
| Vary-Education | 15 vs. 11 | 0.339 | [0.310, 0.369] | 82 | 0.342 | [0.312, 0.372] | 82 |
| Vary-Education | 15 vs. 14 | 0.260 | [0.224, 0.297] | 82 | 0.495 | [0.459, 0.532] | 82 |
| Vary-Education | 15 vs. 15 | 0.081 | [0.052, 0.111] | 82 | 0.512 | [0.493, 0.530] | 82 |
| Vary-Employment | 15 vs. 0 | 0.375 | [0.332, 0.418] | 56 | 0.048 | [0.013, 0.083] | 55 |
| Vary-Employment | 15 vs. 1 | 0.339 | [0.291, 0.387] | 56 | 0.092 | [0.043, 0.140] | 56 |
| Vary-Employment | 15 vs. 4 | 0.298 | [0.249, 0.347] | 56 | 0.214 | [0.150, 0.278] | 56 |
| Vary-Employment | 15 vs. 7 | 0.307 | [0.261, 0.352] | 56 | 0.248 | [0.209, 0.287] | 56 |
| Vary-Employment | 15 vs. 11 | 0.274 | [0.229, 0.318] | 56 | 0.349 | [0.315, 0.383] | 56 |
| Vary-Employment | 15 vs. 14 | 0.220 | [0.172, 0.269] | 56 | 0.471 | [0.437, 0.504] | 56 |
| Vary-Employment | 15 vs. 15 | 0.110 | [0.074, 0.147] | 56 | 0.497 | [0.460, 0.534] | 56 |

*Notes*: Sample sizes for the meritocratic subsample: Vary-Probability: n=170; Vary-Education: n=82; Vary-Employment: n=56.